

# Lab 8: Molecular Genetics — From DNA to Protein

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BIOL-1

Name: \_\_\_\_\_ Date: \_\_\_\_\_

## Learning Objectives

By the end of this lab, you will be able to:

1. Transcribe a DNA sequence into mRNA using base-pairing rules.
2. Translate an mRNA sequence into amino acids using a codon table.
3. Predict the effects of point mutations and frameshift mutations on a protein.
4. Explain the Central Dogma in your own words.

## Overview

Every living cell follows the same basic recipe to turn genetic information into a working molecule:

***DNA → RNA → Protein***

This is the **Central Dogma** of molecular biology. It means your DNA is like a master cookbook that never leaves the kitchen (the nucleus). When a protein needs to be made, the cell copies just the recipe it needs into a messenger (mRNA), which travels to the ribosome — the cell's protein-building machine.

In this lab you will **be the cell**. You'll copy DNA into mRNA (*transcription*), decode mRNA into amino acids (*translation*), and discover what happens when the recipe gets a typo (*mutations*).

## Materials

- This worksheet
  - Codon table (Part 1)
  - Colored pencils (optional — great for color-coding bases!)
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## Part 1: Base Pairing — The Language of DNA and RNA

DNA and RNA are built from four **nucleotide bases**. These bases always pair in specific, predictable ways — like puzzle pieces that only fit together one way. Understanding these pairing rules is the key to everything else in this lab.

### DNA–DNA Pairing (Replication)

When a cell needs to **copy its DNA** before dividing, the double helix unzips and each strand serves as a template. The enzyme **DNA Polymerase** reads the template and adds the matching base according to these rules:

#### Template Base Pairs With

**A** (Adenine)    **T** (Thymine)

**T** (Thymine)    **A** (Adenine)

**G** (Guanine)    **C** (Cytosine)

**C** (Cytosine)    **G** (Guanine)

 **Memory trick:** Always with **T**, Goes with **C** — in DNA, A pairs with T and G pairs with C.

**Quick practice — Write the complementary DNA strand:**

**Template**    **T**    **A**    **C**    **C**    **G**    **T**

New strand    \_\_\_\_    \_\_\_\_    \_\_\_\_    \_\_\_\_    \_\_\_\_    \_\_\_\_

### DNA–RNA Pairing (Transcription)

When the cell needs to **make a protein**, it doesn't send the DNA out of the nucleus. Instead, the enzyme **RNA Polymerase** reads the DNA template and builds a single-stranded **mRNA** (messenger RNA) copy — a process called **transcription**.

RNA has one important difference: it uses **Uracil (U)** instead of Thymine (T).

#### DNA Template Base Pairs With (RNA)

**A** (Adenine)                      **U** (Uracil)

**T** (Thymine)                      **A** (Adenine)

**G** (Guanine)                      **C** (Cytosine)

**C** (Cytosine)                      **G** (Guanine)

 **The only change from DNA→DNA pairing:** Wherever DNA has **A**, the mRNA gets **U** instead of T. Everything else stays the same.

**Quick practice — Transcribe this short DNA template into mRNA:**

DNA Template   T   A   C   C   G   T  
mRNA        \_\_\_\_\_

**Check your understanding:**

**1. What are the two base-pairing rules that are the *same* in both DNA replication and transcription?**

**2. What is the one key difference between DNA–DNA pairing and DNA–RNA pairing?**

**3. If a DNA template strand reads G-A-T-T-C-A, what would the mRNA sequence be?**

## Part 2: Transcription — DNA → mRNA

Now let's apply those pairing rules to a full gene! Transcription is like photocopying one page from a master binder — the original DNA stays safe in the nucleus while the mRNA copy travels to the ribosome.

Use the DNA→RNA pairing rules from Part 1: **A→U, T→A, G→C, C→G**.

**DNA template strand:** 3' — T A C · C G T · A C G · T C G · G G T · G A C · A  
T T — 5'

**4. Transcribe — Write the mRNA sequence. Group into codons (groups of 3).**

**5. Where in a eukaryotic cell does transcription occur?**

**6. What enzyme carries out transcription?**

## Part 3: Translation — mRNA → Protein (The Codon Table)

Once the mRNA reaches the ribosome, it is read in groups of **three bases** called **codons**. Each codon specifies one amino acid (or a stop signal). This process is called **translation** — the cell is literally *translating* the language of nucleotides into the language of amino acids.

### The Codon Table

**How to use it:** Find the first base in the left column, the second base in the top row, and the third base in the right column. The intersection gives you the amino acid.

|   | U                  | C       | A       | G       |                   |
|---|--------------------|---------|---------|---------|-------------------|
|   | <i>2nd base →</i>  |         |         |         | <i>3rd base ↓</i> |
| U | Phe (F)            | Ser (S) | Tyr (Y) | Cys (C) | U                 |
|   | Phe (F)            | Ser (S) | Tyr (Y) | Cys (C) | C                 |
|   | Leu (L)            | Ser (S) | ☹ STOP  | ☹ STOP  | A                 |
|   | Leu (L)            | Ser (S) | ☹ STOP  | Trp (W) | G                 |
| C | Leu (L)            | Pro (P) | His (H) | Arg (R) | U                 |
|   | Leu (L)            | Pro (P) | His (H) | Arg (R) | C                 |
|   | Leu (L)            | Pro (P) | Gln (Q) | Arg (R) | A                 |
|   | Leu (L)            | Pro (P) | Gln (Q) | Arg (R) | G                 |
| A | Ile (I)            | Thr (T) | Asn (N) | Ser (S) | U                 |
|   | Ile (I)            | Thr (T) | Asn (N) | Ser (S) | C                 |
|   | Ile (I)            | Thr (T) | Lys (K) | Arg (R) | A                 |
|   | 🟢 Met (M)<br>START | Thr (T) | Lys (K) | Arg (R) | G                 |
| G | Val (V)            | Ala (A) | Asp (D) | Gly (G) | U                 |
|   | Val (V)            | Ala (A) | Asp (D) | Gly (G) | C                 |
|   | Val (V)            | Ala (A) | Glu (E) | Gly (G) | A                 |
|   | Val (V)            | Ala (A) | Glu (E) | Gly (G) | G                 |

**Key:**

- 🟢 **AUG** = Start codon (also codes for Methionine)

-  **UAA, UAG, UGA** = Stop codons (no amino acid added — translation ends)

## Quick Practice

Try decoding these three codons before moving on:

| Codon | 1st base | 2nd base | 3rd base | Amino Acid |
|-------|----------|----------|----------|------------|
| GCA   | G        | C        | A        |            |
| UUU   | U        | U        | U        |            |
| AUG   | A        | U        | G        |            |

**7. Translate your mRNA from Part 2 — Use the codon table to convert each codon into an amino acid. Write the amino acid chain.**

**8. What was your first codon? What amino acid does it code for, and why is that codon special?**

**9. How did you know to stop translating? What was the stop codon?**

**Bonus look-up:** Find two different codons that both code for Leucine (Leu). Write them here:

This is called **redundancy** — multiple codons can code for the same amino acid. Why might this be useful for an organism? (*Hint: think about mutations.*)



## Part 4: Mutation Detective

Mutations are changes in the DNA sequence — like typos in a recipe. Some are harmless; others can ruin the dish. Let's investigate.

Go back to the **original DNA template strand** from Part 2.

### Scenario A: Substitution (Point Mutation)

The **8th base** (the C in the third codon ACG) changes **C** → **G**.

**Mutated DNA template:** 3' – T A C · C G T · A G G · T C G · G G T · G A C ·  
A T T – 5'

#### 10. Transcribe and translate the mutated sequence.

• Mutated mRNA:

• Amino acid chain:

#### 11. Did the amino acid chain change? Classify the mutation as *silent* (no change), *missense* (different amino acid), or *nonsense* (creates a premature stop).

### Scenario B: Insertion (Frameshift Mutation)

Return to the **original DNA template**. An extra **G** is inserted right after the 3rd base (after the C in TAC).

**Mutated DNA template:** 3' – T A C · G C G · T A C · G T C · G G G · T G A ·  
C A T · T – 5'

#### 12. Transcribe and translate the frameshift-mutated sequence.

• Mutated mRNA:

• Amino acid chain:

**13. Compare the frameshift protein to your original. How many amino acids changed? Why is this type called a "frameshift"?**

### Scenario C: Deletion (Another Frameshift)

Return to the **original DNA template**. This time, **delete the 6th base** (the T in the second codon CGT).

**Mutated DNA template:** 3' – T A C · C G A · C G · T C G · G G T · G A C · A T T – 5'

*(Re-group into codons after the deletion:)* 3' – T A C · C G A · C G T · C G G · G T G · A C A · T T – 5'

**14. Transcribe and translate the deletion-mutated sequence.**

• Mutated mRNA:

• Amino acid chain:

**15. How does a deletion compare to an insertion? Did both cause the same amount of damage to the protein?**

### Comparing All Three Mutations

**16. Rank the three mutations (substitution, insertion, deletion) from least to most damaging. Explain your reasoning.**

## Part 5: Big Picture

**17. Complete the Central Dogma:**

DNA → \_\_\_ → \_\_\_

**18. Where in a eukaryotic cell does transcription occur?**

**19. Where does translation occur?**

**20. What cellular structure reads mRNA and builds proteins (carries out translation)?**

**21. In your own words: How can a single change in someone's DNA affect their physical traits? Trace the path from gene → protein → trait.**

## Part 6: Real-World Connection

**22. Sickle-cell disease** is caused by a single point mutation in the hemoglobin gene. The 6th amino acid changes from Glutamic acid (Glu) to Valine (Val). Based on what you learned today:

- What kind of mutation is this — silent, missense, or nonsense?

- Why can a single amino acid change affect the shape of an entire protein? (*Hint: amino acids fold into 3D shapes, and each one contributes to the fold.*)

**23. Not all mutations are bad!** Can you think of a situation where a mutation might actually help an organism survive? (*Hint: think about how evolution works.*)

## Bonus Challenge

**Design your own mutation!** Start with the original DNA template from Part 2. Create a mutation of your choice (substitution, insertion, or deletion), then transcribe and translate it.

- Type of mutation you chose:
- Your mutated DNA template:
- Mutated mRNA:
- Amino acid chain:
- What happened to the protein? Was it silent, missense, nonsense, or a frameshift?